# PANJAB UNIVERSITY, CHANDIGARH

(Estted. under the Panjab University Act VII of 1947- enacted by the Govt. of India)

## FACULTY OF ARTS

## SYLLABI

***FOR***

## MASTERS IN REMOTE SENSING AND GEOGRAPHIC INFORMATION SYSTEMS (RS & GIS) (SEMESTER SYSTEM) EXAMINATIONS, 2014-2015

**GUIDELINES FOR CONTINUOUS INTERNAL ASSESSMENT (20%) FOR REGULAR STUDENTS OF MASTERS IN REMOTE SENSING AND GEOGRAPHIC INFORMATION SYSTEMS (RS & GIS)**
**(Semester System)**
**(Effective from the First Year Admissions for the Academic Session 2014-2015)**

1. The Syndicate has approved the following guidelines, mode of testing and evaluation including Continuous Internal Assessment of students :

   (i) Terminal Evaluation :80 % (Theory 50% and Practical 30% as specified in syllabus)

   (ii) Continuous Assessment : 20 %

   (iii) Continuous Assessment may include written assignment, snap tests, participation in discussions in the class, term papers, attendance etc.

   (iv) In order to incorporate an element of Continuous Internal Assessment of students, the Colleges/Departments will conduct the following:

| | | | |
|---|---|---|---|
| (a) | Written Test | : | 25 (reduced to 5) |
| (b) | Snap Test | : | 25 (reduced to 5) |
| (c) | Participation in Class discussion | : | 15 (reduced to 3) |
| (d) | Term Paper | : | 25 (reduced to 5) |
| (e) | Attendance | : | 10 (reduced to 02) |

**Total: 100 reduced to 20**

2. Weightage of 2 marks for attendance component out of 20 marks for Continuous Assessment shall be available only to those students who attend 75% and more of classroom lectures/seminars/workshops. The break–up of marks for **attendance component** for theory papers shall be as under :

| Attendance Component | | | Mark/s for Theory Papers |
|---|---|---|---|
| (a) | 75 % and above upto 85 % | : | 1 |
| (b) | Above 85 % | : | 2 |

3. It shall **not be compulsory** to pass in Continuous Internal Assessment. Thus, whatever marks are secured by a student out of 20 marks, will be carried forward and added to his/her score out of 80, i.e. the remaining marks allocated to the particular subject and, thus, he/she shall have to secure pass marks both in the University examinations as well as total of Internal Continuous Assessment and University examinations.

4. Continuous Internal Assessment awards from the affiliated Colleges/Departments must be sent to the Controller of Examinations, by name, **two weeks before** the commencement of the particular examination on the *proforma* obtainable from the Examination Branch.

# PANJAB UNIVERSITY, CHANDIGARH

Outlines of tests, syllabi and courses of reading for Masters in Remote Sensing and Geographic Information Systems (RS & GIS) (Semester System) for the examinations of 2014-15.

| **SEMESTER-I** | | | |
|---|---|---|---|
| **Paper** | **Title** | **Max. Marks** | **Credits** |
| I | Fundamentals of Remote Sensing<br>(a) Theory : 50<br>(b) Practical<br>(Lab work 10 marks+ Record file 10 marks+ Viva 10 marks): 30<br>(c) Internal Assessment : 20 | 100 | 4 |
| II | Fundamentals of Photogrammetry<br>(a) Theory : 50<br>(b) Practical<br>(Lab work 10 marks+ Record file 10 marks+ Viva 10 marks): 30<br>(c) Internal Assessment : 20 | 100 | 4 |
| III | Fundamentals of Geographic Information Systems (GIS)<br>(a) Theory : 50<br>(b) Practical<br>(Lab work 10 marks+ Record file 10 marks+ Viva 10 marks): 30<br>(c) Internal Assessment : 20 | 100 | 4 |
| IV | Fundamentals of Computers and Computer Programming<br>(a) Theory : 50<br>(b) Practical<br>(Lab work 10 marks+ Record file 10 marks+ Viva 10 marks): 30<br>(c) Internal Assessment : 20 | 100 | 4 |
| | **Total** | **400** | **16** |
| **SEMESTER-II** | | | |
| **Paper** | **Title** | **Max. Marks** | **Credits** |
| V | Cartography<br>(a) Theory : 50<br>(b) Practical<br>(Lab work 10 marks+ Record file 10 marks+ Viva 10 marks): 30<br>(c) Internal Assessment : 20 | 100 | 4 |
| VI | Spatial Analysis<br>(a) Theory : 50<br>(b) Practical<br>(Lab work 10 marks+ Record file 10 marks+ Viva 10 marks): 30<br>(c) Internal Assessment : 20 | 100 | 4 |
| VII | Image Processing<br>(a) Theory : 50<br>(b) Practical<br>(Lab work 10 marks+ Record file 10 marks+ Viva 10 marks): 30<br>(c) Internal Assessment : 20 | 100 | 4 |
| VIII | Fundamentals of Global Positioning System (GPS)<br>(a) Theory : 50<br>(b) Practical<br>(Lab work 10 marks+ Record file 10 marks+ Viva 10 marks): 30<br>(c) Internal Assessment : 20 | 100 | 4 |
| | **Total** | **400** | **16** |

**SEMESTER III**

| | | |
|---|---|---|
| Paper VII: | Application of Remote Sensing and GIS in Disaster Management | **200** |
| | (a) Theory : 100 | |
| | (b) Practicals : 60 | |
| | (Lab. work 30 marks+ Record file 20 marks+ Viva 10 marks) | |
| | (c) Internal Assessment : 40 | |
| Paper VIII: | Spatial Analysis | **200** |
| | (a) Theory : 100 | |
| | (b) Practicals : 60 | |
| | (Lab. work 30 marks+ Record file 20 marks+ Viva 10 marks) | |
| | (c) Internal Assessment : 40 | |
| Paper IX: | Research Methodology and Report Writing | **200** |
| | (a) Theory : 100 | |
| | (b) Practicals : 60 | |
| | (Lab. work 30 marks+ Record file 20 marks+ Viva 10 marks) | |
| | (c) Internal Assessment : 40 | |

**SEMESTER –IV**

| | | |
|---|---|---|
| | Project Report | **600** |
| | (a) Mid-Term Appraisal of the Project Report : 100 | |
| | (b) Evaluation of the Project Report : 400 | |
| | (c) Presentation &Viva Voce (50+50) : 100 | |
| | **Grand Total** | **2400** |

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**NOTE:**

1. Those candidates, who wish to discontinue after successful completion of first two semesters of the Masters course shall be awarded Diploma in Remote Sensing and Geographic Information Systems (GIS).
2. The minimum marks required to pass the examination shall be 45% in each (i) theory paper, (ii) practical work, and (iii) field/project report, and 50% in aggregate.
3. The practical examination shall be conducted by a team of three examiners, including the internal, Chairperson of the Department, and one additional faculty member teaching the course.

# SEMESTER - I

## PAPER-I: FUNDAMENTALS OF REMOTE SENSING

**Max. Marks : 100**
Theory : 50
Practical : 30
Internal Assessment : 20

**OBJECTIVES:** To introduce the basic principles of remote sensing, satellite systems and their functioning, and applications of remote sensing technology.

**(A) THEORY** **Marks: 50**
**Time: 3 Hours**

### UNIT – I

Fundamental Concepts of Remote Sensing:

i. Remote Sensing: Definition, Concept, History and Applications
ii. Types of Remote Sensing
iii. Remote Sensing Platforms and Scanning Systems

### UNIT – II

EMR Principles and Interaction Mechanisms:

iv. Radiation Principles; Electromagnetic Spectrum
v. Energy-Atmosphere Interaction; Atmospheric Windows
vi. Energy-Earth Interaction; Spectral Signatures of Surface Features

### UNIT – III

Image Processing and Interpretation:

vii. Image: Meaning and Types (Analogue and Digital) and Characteristics
viii. Resolution: Spatial, Spectral, Radiometric and Temporal
ix. Basics of Image Processing; Elements of Image Interpretation.

### UNIT – IV

Satellite Systems and Sensors:

x. LANDSAT, SPOT and IRS
xi. IKONOS, Quickbird, World-View and GeoEye
xii. SRTM, ASTER, MODIS

**NOTE:**

1. A Compulsory Question at serial number I containing 10 short answer type questions shall be set covering the whole syllabus. Student will attempt any 7 parts in about 25-30 words each. Each part shall carry 2 marks (total 14 marks).
2. A total of eight questions will be set out of the whole syllabus, at least 2 from each unit. The candidates will attempt 4 questions selecting one from each unit carrying 9 marks each (total 36 marks). These will be in addition to the Compulsory Question at serial number I.

**(B) PRACTICAL** **Marks: 30**

1. Comparing Satellite Imageries of Various Spatial Resolution
2. Information on the Margins of a Hard Copy Satellite Image

3. Analysis of Spectral Signatures of Surface Features
4. Visual Interpretation of Features from Satellite Image
5. Digital Satellite Imageries: Data Import & Data Structures
6. Digital Image Interpretation: Band Combination Analysis
7. Digital Image Interpretation: Histogram and Pixel Value Analysis
8. Unsupervised Classification of Satellite Data
9. Supervised Classification of Satellite Data

**NOTE:** The practical examination shall be conducted by a team of three examiners, including the internal, Chairperson of the Department and one additional faculty member teaching the course.

**(C) INTERNAL ASSESSMENT** **Marks: 20**

Internal assessment shall be based on Written Test, Snap Test, Participation in Class discussion, Term Paper and Attendance as prescribed by the University.

## LIST OF READINGS

**Essential Readings:**


1. Campbell, J. B. and R.H. Wynne, *Introduction to Remote Sensing* (Fifth Edition), Guilford Press, 2012.
2. Jensen, J.R., *Introductory Digital Image Processing: A Remote Sensing Perspective*, Pearson Prentice Hall, 2005.
3. Joseph, G. *Fundamentals of Remote Sensing*, Universities Press, 2005.
4. Lillesand, T.M., R.W. Kiefer and J.W. Chipman, *Remote Sensing and Image Interpretation*, 5th Edition, John Wiley and Sons, 2004.
5. Mather, P.M. and M.G. Koch, *Computer Processing of Remotely-Sensed Images: An Introduction,* John Wiley & Sons, 2011.
6. Prost, Gary L., *Remote Sensing for Geoscientists: Image Analysis and Interpretation*, 3rd Edition, CRC Press, Taylor & Francis, 2013.


**Further Readings:**


1. Gomarasca, Mario A, *Basics of Geomatics*, Springer: Heidelberg, 2009.
2. Nag, P. and M. Kudrat, *Digital Remote Sensing*, Concept Publishing Company, New Delhi, 1998.
3. Richards, John A. *Remote Sensing Digital Image Analysis: An Introduction,* 5th Edition, Springer-Verlag Berlin Heidelberg, 2013.
4. Gupta, R.P., *Remote Sensing Geology*, Springer, Heidelberg, Germany, 2003.
5. Roy, P.S.; Van Westen, C.J.; Jha, V.K.; Lakhera, R.C. and Champati Ray, P.K., *Natural Disaster and their Mitigation: Remote Sensing and Geographical Information System Perspectives*, IIRS, Dehra Dun, Govt. of India, 2000.
6. Verbyla, David, L., *Satellite Remote Sensing of Natural Resources*, Lewis Publishers, New York, 2005.


**PEDAGOGY:** Students may be taken to field for verification of identifiable features on satellite imageries; they may be asked to prepare a report using toposheets, satellite imageries and other data sources to trace the temporal and spatial changes. Students may be taken to different institutions to acquaint with different remote sensing equipments, techniques, and products. Remote Sensing professionals may be invited to interact and deliver special lectures to students on remote sensing techniques.

## PAPER II: FUNDAMENTALS OF PHOTOGRAMMETRY

| | |
|---|---|
| **Max. Marks** | **: 100** |
| Theory | : 50 |
| Practical | : 30 |
| Internal Assessment | : 20 |

**Objectives**: The aim of this paper is to provide a general overview of Photogrammetry, its theory and general working principles; to introduce the basic concepts and significance of Photogrammetry and to provide a practical understanding of principles of measurement from vertical aerial photographs.

**(A) THEORY** **Marks: 50**
**Time: 3 Hours**

### UNIT I

i. Photogrammetry: Definition and Categories
ii. Historical Background: Early Developments in Aerial Surveying and Mapping
iii. Problems of Aerial Photogrammetry
iv. Application of Photogrammetry

### UNIT II

v. Basic Mathematics of Photogrammetry: Geometry of Aerial Photographs
vi. Calculation of Photo Scale,
vii. Relief Displacement: Concept and Calculations
viii. Parallax: Concept and Determination
ix. Calculation of Height of Objects on Vertical Aerial Photograph

### UNIT III

x. Stereoscopy: Definition
xi. Principles of vision, Binocular Vision, Stereoscopic Vision Testing,
xii. Pseduoscopic Views
xiii. Types of Stereoscopes and their Operations

### UNIT IV

xiv. Photogrammetry in the Space Age
xv. Photogrammetry for Missiles and Satellite Tracking
xvi. Ortho-photos and Contour Extraction

**NOTE:**

1. A Compulsory Question at serial number I containing 10 short answer type questions shall be set covering the whole syllabus. Student will attempt any 7 parts in about 25-30 words each. Each part shall carry 2 marks (total 14 marks).
2. A total of eight questions will be set out of the whole syllabus, at least 2 from each unit. The candidates will attempt 4 questions selecting one from each unit carrying 9 marks each (total 36 marks). These will be in addition to the Compulsory Question at serial number I.

**(B) PRACTICAL** **Marks: 30**

1. Distinction between Map, Photo and Stereoscopic view.
2. Arrangement of Aerial Photographs with the help of numbers given on the photograph and preparation of Photo Index.

3. Understanding the simple geometry of aerial photographs
4. Interpretation of information given on the margins of photograph and satellite imageries.
5. Locating principal point, conjugate principal point, flight line
6. Stereoscopic vision Test, selection of stereo pair and its orientation and placing under the stereoscope for stereovision.
7. Identification and mapping of features from single photograph.
8. Identification and mapping of features from a stereo pair and their verification in the field.
9. Measurement of scale, relief displacement , parallax , height and slope from vertical aerial photographs
10. Identification of planimetric positions

**NOTE:** The practical examination shall be conducted by a team of three examiners, including the internal, Chairperson of the Department and one additional faculty member teaching the course.

**(C) INTERNAL ASSESSMENT** **Marks: 20**

Internal assessment shall be based on Written Test, Snap Test, Participation in Class discussion, Term Paper and Attendance as prescribed by the University.

## LIST OF READINGS

**Essential Readings**


1. McGlone, J. Chris, *Manual of Photogrammetry (Sixth Edition),* American Society of Photogrammetry and Remote Sensing, 2013.
2. Mikhail, Edward M. et al, *Introduction to Modern Photogrammetry*, John Wiley & Sons Inc. 2001.
3. Moffitt, F. H., *Photogrammetry*, 3rd Edition, Harper & Row, NY, 1980.
4. Paul Wolf et al., *Elements of Photogrammetry with Application in GIS*, Fourth Edition, McGraw-Hill Professional, 2013.
5. Wolf, P. R., *Elements of Photogrammetry*. McGraw-Hill, NY, 1983.
6. Zorn, H. C., *Introductory Course in Photogrammetry*. 6th Ed. ITC, Netherlands, 1980.


**Further Readings**


1. Burnside, C. D., *Mapping from Aerial Photography*. 2nd Ed, Collins, 1985.
2. Campbell, J.B., *Introduction to Remote Sensing*, 3rd ed., The Guilford Press, 2002.
3. Curran Paul, J., *Principles of Remote Sensing*, UK: ELBS, 1984.
4. Joseph, George, *Fundamentals of Remote Sensing,* Universities Press India, 2007.
5. Lillesand, T. M.; R.W. Kiefer, and J.W. Chipman, *Remote Sensing and Image Interpretation*, 5th Edition, Wiley, 2007.
6. Sabins, Floyd F., *Remote Sensing: Principles and Interpretation,* New York: WH Freeman and Company, 2007.


**PEDAGOGY:** Students will be demonstrated the difference between qualitative and quantitative understanding of various aspects of Photogrammetry. Demo exercise on measurements on the map and photo will make students aware the importance of Photogrammetry in objective and precise mapping using aerial photographs.

## PAPER III: FUNDAMENTALS OF GEOGRAPHIC INFORMATION SYSTEMS (GIS)

| | |
|---|---|
| **Max. Marks** | **: 100** |
| Theory | : 50 |
| Practical | : 30 |
| Internal Assessment | : 20 |

**OBJECTIVES:** The main objective of this course is to expose the students to fundamental principles of Geographical Information Systems including basic concepts and definitions, methods and techniques.

**(A) THEORY** **Marks: 50**
**Time: 3 hours**

### UNIT – I

Concepts and Definitions:

i. Geographic Information Systems (GIS): Definition and Developments
ii. Spatial data base: Points, Lines and Polygons
iii. GIS Components: Hardware and Software

### UNIT – II

Functional Elements:

iv. Data Acquisition, Input and Editing
v. Data Manipulation and Topology Creation
vi. Data Analysis and Query

### UNIT – III

Data Management and Structure:

vii. Data Base Management System (DBMS): Purpose and structure
viii. Classification: Raster, Vector, Quadtree, Hierarchical, Network and Relational

### UNIT - IV

GIS Applications:

ix. GIS Applications
x. GIS: Decision Support System

**NOTE:**

1. A Compulsory Question at serial number I containing 10 short answer type questions shall be set covering the whole syllabus. Student will attempt any 7 parts in about 25-30 words each. Each part shall carry 2 marks (total 14 marks).
2. A total of eight questions will be set out of the whole syllabus, at least 2 from each unit. The candidates will attempt 4 questions selecting one from each unit carrying 9 marks each (total 36 marks). These will be in addition to the Compulsory Question at serial number I.

**(B) PRACTICAL** **Marks: 30**

1. GIS Software: Introduction to Arc GIS
2. Georeferencing Maps/Images
3. Digitization of Raster Map: Point, Line and Polygon Features
4. Preparation of Attribute Tables, Editing and Joining Tables
5. Analyzing Attribute Data: Calculating Area, Perimeter, Length

6. Spatial Representation: Mapping Techniques
7. Spatial Representation: Symbolizing and Map Layouts
8. Basic Analysis in GIS: Buffering, Overlay and Query Building

**NOTE:** The practical examination shall be conducted by a team of three examiners, including the internal, Chairperson of the Department and one additional faculty member teaching the course.

**(C) INTERNAL ASSESSMENT** **Marks: 20**

Internal assessment shall be based on Written Test, Snap Test, Participation in Class discussion, Term Paper and Attendance as prescribed by the University.

## LIST OF READINGS

**Essential Readings:**


1. Harvey, F., *A Primer of GIS: Fundamental Geographic and Cartographic Concepts*, Guilford Press, New York, 2008.
2. Ian Heywood,Sarah Cornelius and Steve Carver, *An Introduction to Geographical Information Systems*, 3rd Edition, Pearson, Dorling Kindersley (India) Pvy. Ltd, 2010.
3. Kang-tsung (Karl) Chang, *Introduction to Geographic Information Systems*, 4th Edition, Tata McGraw-Hill, New Delhi, 2009.
4. Maguire, D.J.; M. Batty and M.F. Goodchild, *GIS, Spatial Analysis and Modeling*, ESRI Press, 2005.
5. Michael F. Goodchild, *Introduction to GIS*, Santa Barbara, California, & Karan K. Kemp (eds.) NGGIA, 1990.
6. Muralikrishna, I.V., *Remote Sensing Applications and Geographic Information Systems - Recent Trends*, Tata McGraw Hill, New Delhi, 1992.
7. Wise, S., *GIS Fundamentals*, 2nd Edition, CRC Press, Taylor & Francis Group, 2013.


**Further Readings:**


1. Aronoff, S., *Geographic Information Systems: A Management Perspective***,** WDL Publications Ottawa,Canada, 1992.
2. Burrough, P.A. & Macdoneli, R.A., *Principles of Geographic Information Systems,* Oxford University Press, 2000.
3. Demers, M.N., *Fundamentals of Geographic Information Systems*, 2nd Edition, John Wiley and sons, 2000.
4. Jefrey S. & John, E., *Geographic Information Systems - An Introduction*, Prentice Hall, New Jersey, USA, 1990.
5. Laurini, R. & Thomson, D., *Fundamentals of Spatial Information Systems*, Academic Press, London, 1994.
6. Rashid, S.M., *Remote Sensing in Geography*, Manak Publications, Delhi, 1993.
7. Shen, Zhenjiang, *Geospatial Techniques in Urban Planning*, Springer-Verlag Berlin Heidelberg, 2012.


**PEDAGOGY:** Taking students for fieldwork, inviting professionals for lectures, taking students to institutes/training centers engaged in imparting professional training and providing research consultancy in business, tourism and area/regional planning will be the main component of the teaching methodology.

## PAPER-IV: FUNDAMENTALS OF COMPUTERS AND COMPUTER PROGRAMMING

**Max. Marks : 100**
Theory : 50
Practical : 30
Internal Assessment : 20

**OBJECTIVES**: To develop basic skills and understanding of computer operations; to develop skills of Word Processing for written communication of results and documentation of work performed; to impart skills for data management, charting, and visualization; to expose students to the effective techniques of using search engines & the services of Internet and to impart programming skill for developing programs in 'C' language.

**(A) THEORY** **Marks: 50**
**Time: 3 Hours**

### UNIT-I

Computer Concepts:
i. Introduction to Computers
ii. Computer System, Hardware, Software, Input and Output Devices
iii. Data Representation
iv. Fundamental Concepts of Operating Systems.

OS and Application Software:
v. Introduction to Windows Operating System.
vi. Introduction to MS-Word and MS-Excel Software.

### UNIT-II

Computer Graphics:
vii. Fundamentals of Computer Graphics, Image Formats

Computer Networks:
viii. Introduction to computer networks
ix. Overview of Internet and its services,
x. Search Engines, Techniques to search effectively.

Web Application Development:
xi. Introduction to web application development using HTML.

### UNIT-III

Data concepts:
xii. Key Database Concepts
xiii. Database Management Systems, Type of DBMS
xiv. Structured Query Language (SQL)
xv. Data Visualization, Data Mining.

Information Systems
xvi. Introduction to Information Systems
xvii. Types of Information Systems.

### UNIT-IV

Programming Concepts:
xviii. Concept of a program, data types, keywords, structure of a program.

Writing 'C' programs:
xix. Statements, Types of Statements
xx. Library Functions and User Defined Functions
xxi. Arrays
xxii. File Handling.

**NOTE:**

1. A Compulsory Question at serial number I containing 10 short answer type questions shall be set covering the whole syllabus. Student will attempt any 7 parts in about 25-30 words each. Each part shall carry 2 marks (total 14 marks).
2. A total of eight questions will be set out of the whole syllabus, at least 2 from each unit. The candidates will attempt 4 questions selecting one from each unit carrying 9 marks each (total 36 marks). These will be in addition to the Compulsory Question at serial number I.

**(B) PRACTICAL** **Marks: 30**

1. Hands on experience on Windows operating system.
2. Hands on experience on MS-Word
3. Hands on experience on MS-Excel
4. Creating web pages using HTML,
5. Techniques to use search engines effectively.
6. Writing queries using SQL.
7. Writing C programs.

**Note:** The practical examination shall be conducted by a team of three examiners, including the internal, Chairperson of the department, and one additional faculty member teaching the course.

**(C) INTERNAL ASSESSMENT** **Marks: 20**

Internal assessment shall be based on Written Test, Snap Test, Participation in Class discussion, Term Paper and Attendance as prescribed by the University.

## LIST OF READINGS

**Essential Readings**


1. Miller, M., *Easy Computer Basics, Windows 8.1 Edition*, Pearson Education, Inc. 2014.
2. Basandra, S. K., *Computers Today*, Galgotia Publications Pvt. Ltd., 2010.
3. Salaria, R.S. *Computer Concepts and Programming in C*, Salaria Publishing House, 2013.
4. Lambert, J. and J. Cox, *Microsoft Word 2013: Step by Step*, Eastern Economy Edition, 2013.
5. Frye, C., *Microsoft Excel 2013: Step by Step*, Microsoft Press, 2013.
6. Online Resources: www.w3schools.com for HTML and SQL. (*Live and latest).*


**Further Readings**


1. Freedman, J., *Microsoft Word 2013: Plain & Simple*, O'Reilly Media Inc., 2013.
2. Norton, P., *Introduction to Computers*, 6th Edition, McGraw-Hill Education, 2004.
3. Norton, P., *Computing Fundamentals*, Student Edition, McGraw-Hill, 2004.


**PEDAGOGY:** Bottom up approach of explaining concepts; programming exercises to simulate interest to practice programming applications; Audio visual presentations; Test/Assignment as per the latest in industry/real-life. Students will be encouraged to ask questions and participate in the classroom discussion.

# SEMESTER-II

## PAPER V: CARTOGRAPHY

**Max. Marks : 100**
Theory : 50
Practical : 30
Internal Assessment : 20

**OBJECTIVES:** The course is designed to emphasize the relevance and scope of cartography in GIS. It shall highlight the value of this vital field in the current and highly versatile field of Remote Sensing and GIS.

**(A) THEORY** **Marks: 50**
**Time: 3 Hours**

### UNIT – I

Basic Concepts of Cartography

i. Cartography: Definition, Scope and Significance
ii. Historical Perspective of Cartographic Development
iii. Traditional and Digital Cartography
iv. Cartography and GIS

### UNIT – II

Data Types and Map Basics

v. Levels of Measurements
vi. Data: Types (Spatial and Attribute data), Sources and Analysis
vii. Map Types and Classifications; Topographic Maps
viii. Map Projections, Datum and Coordinate System

### UNIT – III

Map Designs and Production

ix. Principles of Symbolization
x. Basics of Colour Schemes and Representation
xi. Map Lettering and Legend Presentation
xii. Map Layout and Product Generation

### UNIT – IV

Interpretation of Topographical Maps

xiii. Relief Representation: Contours, Hachures, Hill Shading
xiv. Relief (Absolute and Relative) and Profiles (Longitudinal and Transverse)
xv. Slope Analysis (Wentworth and Robinson)
xvi. Reading Topographical Maps

**NOTE:**

1. A Compulsory Question at serial number I containing 10 short answer type questions shall be set covering the whole syllabus. Student will attempt any 7 parts in about 25-30 words each. Each part shall carry 2 marks (total 14 marks).
2. A total of eight questions will be set out of the whole syllabus, at least 2 from each unit. The candidates will attempt 4 questions selecting one from each unit carrying 9 marks each (total 36 marks). These will be in addition to the Compulsory Question at serial number I.

**(B) PRACTICAL** **Marks: 30**

1. Classification of Topographical Sheets
2. Slope Analysis (Wentworth and Robinson)

3. Generation of Hill Shading Map
4. Profiles: Longitudinal and Transverse Profiles
5. Interpretation of Physical and Cultural features from Topographical Sheet
6. Thematic Maps: Dot Map, Choropleth Map, Proportionate Circles, Isopleth Map

**NOTE:** The practical examination shall be conducted by a team of three examiners, including the internal, Chairperson of the Department and one additional faculty member teaching the course.

**(C) INTERNAL ASSESSMENT** **Marks: 20**

Internal assessment shall be based on Written Test, Snap Test, Participation in Class discussion, Term Paper and Attendance as prescribed by the University.

## LIST OF READINGS

**Essential Readings**


1. Cromley, R.G., *Digital Cartography*, Prentice-Hall of India, New Delhi, 1992.
2. Crampton, W.C., *Mapping: A Critical Introduction to Cartography and GIS,* John Willy & Sons, New York, 2010.
3. Dent, B.D., *Cartography-Thematic Map Design*, WBC McGraw–Hill, Boston, 1999.
4. John Krygier and Denis Wood, *Making Maps: A visual guide to map design for GIS,* Guilford Publications, New York, 2005.
5. Keates, J.S., *Cartographic Design and Production*, Longman, London, 1998.
6. Misra, R.P. and Ramesh, A., *Fundamental of Cartography*, Concept Publishing Company, New Delhi, 1989.
7. Monkhouse, F.J., *Maps and Diagrams*, Methuen and Co., London, 1994.
8. Nag, P., *Thematic Cartography and Remote Sensing*, Concept Publishing Co., New Delhi ,1992.
9. Nelson Petrie*, Analysis and Interpretation of Topographical Maps (Rev),* Orient Blackswan Pvt. Ltd. 1997.
10. Rampal, K.K., *Mapping and Compilation*, Concept Publishing Co., New Delhi, 1993.
11. Raisz, Erwin, *Principles of Cartography*, McGraw Hill, New York, 1962.
12. Robinson, A.H. and others, *Elements of Cartography*, John Willy & Sons, New York, 6th edition, 2009.
13. Singh, R.L. and Singh R.P.B., *Elements of Practical Geography,* Kalyani Publishers , New Delhi, Reprint 2002.


**Further Readings:**


1. Birch, T., *Maps- Topographical and Statistical*, Clarendon Press, Oxford, 1949.
2. Brown, L.A., *The Story of Maps,* Cressit Press, London, 1951.
3. Harley J.B. and David Woodward, *The History of Cartography* Volume 2, Book 2, The University of Chicago Press, 1995.
4. Raisz, Erwin, *Principles of Cartography*, McGraw Hill, New York, 1962.
5. Ramamurthy, K., *Maps Interpretation: India Landscapes Through Survey of India*, Topographic Maps, R.K. Mutt Road, Madras, 1982.
6. Tyner, J.A., *Principles of Map Design*, Guilford Publications, New York, 2010.
7. Wood Clifford H. and Keller C., *Cartographic Design- Theoretical and Practical Perspectives*, John Wiley & Sons, 1996.


**PEDAGOGY:** Students will be encouraged to ask questions and participate in classroom discussions. Assignments, term papers and group discussions will be the additional methods to make teaching more focused and interesting.

## PAPER-VI: SPATIAL ANALYSIS

**Max. Marks : 100**
Theory : 50
Practical : 30
Internal Assessment : 20

**OBJECTIVES:** The course is designed to help the students to understand the techniques, which assist in providing focused and precise analysis of emerged pattern of geographical phenomena.

**(A) THEORY** **Marks: 50**
**Time: 3 Hours**

### UNIT- I

Fundamentals of Spatial analysis

i. Significance of Spatial analysis
ii. Nature of Spatial Data
iii. Using GIS for Spatial Analysis.

### UNIT- II

Point Pattern Analysis

iv. Point Features Over Space: Meaning and Nature
v. Measures of Central tendency: Mean Centre, Weighted Mean Centre, Median Centre
vi. Measures of Dispersion: Standard distance, Standard Deviation Ellipse, Nearest Neighbour Analysis
vii. Spatial Autocorrelation

### UNIT –III

Line Pattern Analysis

viii. Linear Features Over Space: Meaning and Nature
ix. Network Analysis: Connectivity or Topology

Polygon Pattern Analysis

x. The Meaning of Spatial Relationships and Spatial Dependency
xi. Spatial Autocorrelation

### UNIT-IV

Vector Analysis

xii. Buffering: Point, Line, Area buffers
xiii. Overlay: Union, Intersect, Symmetrical Difference, Identity, Point-in-Polygon, Line-in-Polygon, Polygon-in-Polygon

Surface Analysis

xiv. Digital Elevation Model (DEM) and Digital Terrain Model (DTM)

**NOTE:**

1. A Compulsory Question at serial number I containing 10 short answer type questions shall be set covering the whole syllabus. Student will attempt any 7 parts in about 25-30 words each. Each part shall carry 2 marks (total 14 marks).
2. A total of eight questions will be set out of the whole syllabus, at least 2 from each unit. The candidates will attempt 4 questions selecting one from each unit carrying 9 marks each (total 36 marks). These will be in addition to the Compulsory Question at serial number I.

**(B) PRACTICAL** **Marks: 30**

1. Familiarization with Spatial Analysis Software.
2. Measuring Mean Centre, Weighted Mean Centre and Median Centre
3. Measuring Standard Distance and Standard Distance Ellipse
4. Nearest Neighbour Analysis
5. Spatial Autocorrelation of Point Features
6. Network Analysis
7. Vector Analysis: Buffering, Overlay and Query
8. Surface Analysis: Digital Elevation Model (DEM), Digital Terrain Model (DTM)

**NOTE:** The practical examination shall be conducted by a team of three examiners, including the internal, Chairperson of the Department and one additional faculty member teaching the course.

**(C) INTERNAL ASSESSMENT** **Marks: 20**

Internal assessment shall be based on Written Test, Snap Test, Participation in Class discussion, Term Paper and Attendance as prescribed by the University.

## LIST OF READINGS

**Essential Readings:**


1. Chang, Kang-tsung, *Introduction to Geographic Information Systems,* Sixth Edition, Tata McGraw-Hill, 2011.
2. Fotheringham Stewart A and Rogerson A. Peter, *The SAGE Handbook of Spatial Analysis*, SAGE Publications, London, 2009.
3. Laurini, R. & Thomson, D., *Fundamentals of Spatial Information Systems*, Academic Press, 1994.
4. Sanders, L., *Models in Spatial Analysis*, John Wiley & Sons, 2013.
5. Wong, David W.S. & Jay Lee, *Statistical Analysis of Geographic Information with ArcView GIS and ArcGIS*, John Wiley and Sons, 2005.


**Further Readings:**


1. Burrough, P.A. & Macdoneli, R.A., *Principles of Geographic Information Systems,* Oxford University Press, 2000.
2. Demers, M.N., *Fundamentals of Geographic Information Systems*, 2nd Edition, John Wiley and sons, 2000.
3. Maguire, D.J.; M. Batty and M.F. Goodchild, *GIS, Spatial Analysis and Modeling*, ESRI Press, 2005.
4. Murayama, Y. and R. Thapa, *Spatial Analysis and Modeling in Geographical Transformation Process: GIS Based Applications*, Springer, 2011.


**PEDAGOGY:** Students may be asked to prepare a report on spatial analysis applications using external and self generated databases. Students will be encouraged to ask questions and participate in classroom discussions.

## PAPER VII: IMAGE PROCESSING

| | |
|---|---|
| **Max. Marks** | **: 100** |
| Theory | : 50 |
| Practical | : 30 |
| Internal Assessment | : 20 |

**OBJECTIVES:**

The course is designed to introduce the student to analytical tools and methods which are currently used in digital image processing as applied to image information for human viewing. Then apply these tools in the laboratory in image restoration, enhancement and compression.

**(A) THEORY** **Marks: 50**

**Time: 3 Hours**

### UNIT – I

i. Digital Image Pre-processing: Sources of Errors, Types of Error- Radiometric and Geometric

ii. Image Rectification: Geometric Correction, Radiometric Correction, and Noise Removal.

### UNIT - II

iii. Image Enhancement Techniques: Contrast Enhancement; Linear Contrast Stretch, Histogram Equalisation, Gaussian Stretch, Density Slicing, Pseudocolour Transform.

### UNIT - III

iv. Image Transformation: Band Ratioing, Principal Component Analysis, Hue, Saturation and Intensity Transformation.

v. Spatial Filtering: Low Frequency and High Frequency, Edge Enhancement.

### UNIT - IV

vi. Digital Image Classification: Supervised; Unsupervised; Accuracy Assessment.

**NOTE:**

1. A Compulsory Question at serial number I containing 10 short answer type questions shall be set covering the whole syllabus. Student will attempt any 7 parts in about 25-30 words each. Each part shall carry 2 marks (total 14 marks).
2. A total of eight questions will be set out of the whole syllabus, at least 2 from each unit. The candidates will attempt 4 questions selecting one from each unit carrying 9 marks each (total 36 marks). These will be in addition to the Compulsory Question at serial number I.

**(B) PRACTICAL** **Marks: 30**

1. Familiarization with Image Processing Software
2. Georeferencing/Image Registration
3. Layer Stacking
4. Image Mosaic Preparation & Image Subset
5. Image Enhancement & Transformation Techniques
6. Filtering Techniques
7. Supervised Classification
8. Unsupervised Classification
9. Accuracy Assessment

**NOTE:** The practical examination shall be conducted by a team of three examiners, including the internal, Chairperson of the Department and one additional faculty member teaching the course.

**(C) INTERNAL ASSESSMENT** **Marks: 20**

Internal assessment shall be based on Written Test, Snap Test, Participation in Class discussion, Term Paper and Attendance as prescribed by the University.

## LIST OF READINGS

**Essential Readings:**


1. Burger W., & Burge, M.J., *Principles of Digital Image Processing,* Springer-Verlag London Limited, 2009.
2. Campbell, J. B. and R.H. Wynne, *Introduction to Remote Sensing* (Fifth Edition), Guilford Press, 2012.
3. Chen, C.H., *Image Processing for Remote Sensing,* CRC Press, 2008.
4. Gonzalez, R.C. & Woods, R.E., *Digital Image Processing*, Second edition, Prentice-Hall, Inc., 2002.
5. Jensen, J.R., *Introductory Digital Image Processing: A Remote Sensing Perspective*, Prentice Hall, New Jersey, 1986.
6. Liu, J.G. & Mason, P.J., *Essential Image Processing and GIS for Remote Sensing,* John Wiley & Sons, 2009.
7. Mather, P.M. & Magaly, K., *Computer Processing of Remotely-Sensed Images: An Introduction,* Fourth Edition, John Wiley & Sons, 2011.
8. Richards, John A. *Remote Sensing Digital Image Analysis: An Introduction,* 5th Edition, Springer-Verlag Berlin Heidelberg, 2013.


**Further Readings:**


1. Gomarasca, Mario A, *Basics of Geomatics*, Springer: Heidelberg, 2009.
2. Gupta, R.P., *Remote Sensing Geology*, Springer, Heidelberg, Germany, 2003.
3. Jain, A.K., *Fundamentals of Digital Image Processing,* Prentice-Hall, Inc., 2002.
4. Joseph, G. *Fundamentals of Remote Sensing*, Universities Press, 2005.
5. Lillesand, T.M., Kiefer, R.W. & Chipman, J. W., *Remote Sensing and Image Interpretation*, Fourth Edition, John Wiley and Sons, 2004.
6. Pratt, W.K., *Digital Image Processing*, Third Edition, John Wiley & Sons, 2001.
7. Prithvish Nag and M. Kudrat, Digital *Remote Sensing*, Concept Publishing Company, New Delhi, 1998.
8. Prost, Gary L., *Remote Sensing for Geoscientists: Image Analysis and Interpretation*, 3rd Edition, CRC Press, Taylor & Francis, 2013.
9. Verbyla, David, L., *Satellite Remote Sensing of Natural Resources*, Lewis Publishers, New York, 2005.


**PEDAGOGY:** Students may be asked to prepare a report on image processing techniques using satellite images. Students will be encouraged to ask questions and participate in classroom discussions.

## PAPER VIII: FUNDAMENTALS OF GLOBAL POSITIONING SYSTEM (GPS)

| | |
|---|---|
| **Max. Marks** | **: 100** |
| Theory | : 50 |
| Practical | : 30 |
| Internal Assessment | : 20 |

**Objectives**: The objectives of this paper are to give a comprehensive introduction to GPS and DGPS technology, system concepts, design, operation, implementation and applications, to provide detailed information on the GPS signal, the processing by the receiver, and the techniques by which GPS obtains position, velocity and time.

**(A) THEORY** **Marks: 50**
**Time: 3 Hours**

### UNIT I

i. Concept and Characteristics of Global Navigational Satellite System (GNSS)
ii. Historical Background, GPS Segments, Features of GPS Satellites, Principles of Operation, Fundamentals on Satellite Geodesy, Referential Systems related to GNSS, Survey and Characteristics of Observed Values.

### UNIT II

iii. Surveying with GPS: Absolute, Relative and Differential Positioning, Kinematic GPS.
iv. Performing GNSS Measuring and Data Processing, Overview on Geodetic Data and Data Transformation

### UNIT III

v. Survey on GNSS Application: Basic Principles in Working with DGNSS-Mathematical Models, Coordinate Systems in Function, Practical Problems-Navigation using GNSS
vi. Methods for Determining and Techniques for Searching Ambiguities both for Phase Data and the Combination of Code and Phase Data GNSS Application in: Geodesy, Geodynamics, Tectonic Plate Movement with Time Alternating Coordinates, Navigation, Space.

### UNIT IV

vii. Diverse Systems for Global Positioning: TRANST, DORIS, GLONASS, GPS, GALILEO, GAGAN.
viii. Location-based Services: Practical Application.

**NOTE:**

1. A Compulsory Question at serial number I containing 10 short answer type questions shall be set covering the whole syllabus. Student will attempt any 7 parts in about 25-30 words each. Each part shall carry 2 marks (total 14 marks).
2. A total of eight questions will be set out of the whole syllabus, at least 2 from each unit. The candidates will attempt 4 questions selecting one from each unit carrying 9 marks each (total 36 marks). These will be in addition to the Compulsory Question at serial number I.

**(B) PRACTICAL** **30 Marks**

1. Components and Operation of GPS Instrument
2. Understanding the Different Pages of GPS
3. Collecting Readings using Primary Pages
4. Surveying a Small Area
5. Mapping of the Surveyed Area
6. Tabulation of Information of the Surveyed Area
7. Analysis of the Map and Data Collected
8. Application Exercises

**NOTE:** The practical examination shall be conducted by a team of three examiners, including the internal, Chairperson of the Department and one additional faculty member teaching the course.

**(C) INTERNAL ASSESSMENT** **Marks: 20**

Internal assessment shall be based on Written Test, Snap Test, Participation in Class discussion, Term Paper and Attendance as prescribed by the University.

## LIST OF READINGS

**Essential Readings:**


1. Anderle, R., *The Global Positioning System*, Royal Society of London, UK, 1988.
2. Blewitt, G., *Advances in GPS Technology for Geodynamics Investigations: 1978-1992*, Crustal Dynamics Project AGU Monograph, NASA. 1993.
3. Colombo O. L. & M. M. Watkins, *Satellite Positioning*, in US National report to IUGC on Geodesy, 1991.
4. C. Jones, *Geographical Information Systems and Computer Cartography*, Pearson Education Inc, 1997.
5. C. Rizos, *Introduction to GPS*, University of New South Wales, 1999.
6. El-Rabbani, Ahmed, *Introduction to GPS: The Global Positioning System*, Artech House, 2002.
7. Kaplan, Elliott D., *Understanding GPS: Principles and Applications*, Artech House, 2005.
8. Letham, Lawrence, *GPS Made Easy: Using Global Positioning Systems in the Outdoors*, Rocky Mountain Books, 2008.


**Further Readings:**


1. Goad, C. C., *Proceedings of First International Symposium on GPS*, USA, 1985.
2. Fell, P. J., *Geodetic Positioning Using GPS*, The Ohio State University Report no. 299, Columbus, Ohio, USA, 1980.
3. King, R. W. & Others, *Surveying GPS*, University of New South Wales, Australia, 1985.


**PEDAGOGY:** Students will be provided demo exercises to increase their awareness about wide ranging applications of GPS.

# SEMESTER III

**PAPER VII: APPLICATION OF REMOTE SENSING AND GIS IN DISASTER MANAGEMENT** **Marks: 200**

**(a) THEORY** **Marks: 100**
**Time: 3 Hours**

**OBJECTIVES:**
Main objectives of this course are to introduce the students with the nature, causes and effects of disasters and applications of GIS and Remote Sensing in mapping, mitigation and monitoring of hazards and disasters.

**UNIT – I**

i. Fundamental concepts of hazards and disasters
ii. Natural and human induced disasters: classification, types
iii. Historical perspective of disasters in India

**UNIT – II**

iv. Disaster management: concepts relating to the pre-disaster phase, emergency phase, post-disaster phase
v. Disaster management mechanism in India: agencies, plans and policies
vi. Role of remote sensing and GIS in disaster management

**UNIT – III**

vii. Remote sensing and GIS application in geological hazards: Landslides and Earthquakes.
viii. Remote sensing and GIS application in Hydro meteorological Hazards: Floods, Cyclones, Drought.

**UNIT – IV**

ix. Remote sensing and GIS application in Environmental hazards: Deforestation, Land Degradation and desertification, Forest fires
x. Case Studies: Earthquakes in India, Drought in India, Floods in Indo-Gangetic plains, Landslides in Himalayan region,

**NOTE:**
1. A compulsory question containing 15 short answer type questions shall be set covering the whole syllabus. Student will attempt any 10 parts in about 25-30 words each. Each part shall carry 2 marks (total 20 marks).
2. A total of eight questions will be set out of the whole syllabus, at least 2 from each unit, and the candidates will attempt 4 questions selecting one from each unit, in addition to the compulsory question at serial number I.

**(b) PRACTICAL** **Marks: 60**

1. Major Disaster Events in India (1900-2004)
2. Earthquake Prone Districts in India
3. Cyclone Prone Districts in India
4. Flood Prone Districts in India
5. Northwest Himalayas: Landslides Analysis
6. Himachal Pradesh: Earthquake Analysis

**NOTE:** The practical examination shall be conducted by a team of three examiners, including the internal, Chairperson of the Department and one additional faculty member teaching the course.

**(c) INTERNAL ASSESSMENT** **Marks: 40**

Internal assessment shall be based on Written Test, Snap Test, Participation in Class discussion, Term Paper and Attendance as prescribed by the university.

**BOOKS RECOMMENDED**

**ESSENTIAL READINGS**


| | | |
|---|---|---|
| Greg Bankoff , Georg Frerks, Dorothea Hilhorst | : | *Mapping Vulnerability: Disasters, Development and People*, Earthscan Publications Ltd., 2004. |
| Monmonier mark, | : | *Cartographies of Danger: Mapping Hazards in America***,** University of Chicago Press, 1997. |
| Roy, P.S.; Van Westen, C.J.; Jha, V.K.; Lakhera, R.C. and Champati Ray, P.K., | : | *Natural Disaster and their Mitigation: Remote Sensing and Geographical Information System Perspectives*, IIRS, DehraDun, Govt. of India, India, 2000. |


**FURTHER READINGS**


| | | |
|---|---|---|
| Agarwal, C.S. and P.K. Garg | : | *Text Book on Remote Sensing in Natural Resources Monitoring and Management*, Wheeler Publishing Co, New Delhi, 2000. |
| Burrough, P.A. & Macdoneli, R.A. | : | *Principles of Geographic Information Systems,* Oxford University Press 2000. |
| Demers, M.N. | : | *Fundamentals of geographic Information Systems*, 2nd Edition, John Wiley and sons 2000. |
| Giarratano, J. & Rieley, G. | : | *Expert Systems: Principles and Programming*, PRS Kent Publishing Company 1985. |
| Muralikrishna, I.V. | : | *Remote Sensing Applications and Geographic Information Systems Recent Trends*, Tata McGraw Hill, New Delhi, 1992. |
| Rashid, S.K. | : | *Remote Sensing in Geography*, Manak Publications, Delhi, 1993. |
| Rees, W.G. | : | *Physical Principles of Remote Sensing*, Cambridge University Press, Cambridge, 1990. |
| Swain, Phillip, H. | : | *Remote Sensing*: The quantitative Approach, McGraw Hill International Book Co., 1978. |
| Verbyla, David, L | : | *Satellite Remote Sensing of Natural Resources*, Lewis Publishers New York, 2005. |


**PEDAGOGY:**

Students may be asked to prepare a report on zonation of hazards and disasters using topographical sheets, aerial photographs and satellite images. Students may acquaint with the satellite imageries of various kinds of disasters.

**PAPER VIII: SPATIAL ANALYSIS** **Marks: 200**

**(a) THEORY** **Marks: 100**
**Time: 3 Hours**

**OBJECTIVES:**
The course is designed to introduce the student to spatial analysis capabilities of GIS. To understand the logical connections between attribute data and map features, and the operational procedures built on the spatial relationships among map features.

**UNIT - I**

Introduction to Spatial analysis:

- Significance of Spatial analysis.
- Using GIS for Spatial Analysis.
- Quantification of Spatial Analysis
- Single Layer operations
- Multiple layer operations

**UNIT - II**

Vector Based Spatial Data Analysis

- Overlay
  - Point-in-Polygon, Line-in-Polygon, Polygon-in-Polygon
- Logical operators and Conditional Operators
- Types of Spatial Operations in GIS
  - Identity, Intersect, Union
  - Clip, Erase, Reselect
- Buffering: Point, Line, Area buffers

Raster Based Spatial Data Analysis

- Local functions
- Focal functions
- Zonal Functions
- Global functions

**UNIT - III**

Point Pattern Analysis

- Descriptive Statistics of Point Features
- Spatial Arrangement
  - Nearest Neighbour Index
  - Quadrat Analysis,
- Spatial Autocorrelation

Network Analysis

- Network Data Model
- Path Determination
- Resource allocation/distribution analysis
- Utility Locating/ Siting analysis
- Shortest path

**UNIT - IV**

Spatial Modeling

- Model building
- Multiple Regression
- Logistic Regression
- Data Manipulation and Programming

Surface Analysis

- Digital Elevation Model (DEM) and Digital Terrain Model (DTM)
- Contouring, Hill Shading and Perspective View (3D viewing)
- Slope and Aspect Analysis

**NOTE:**

1. A compulsory question containing 15 short answer type questions shall be set covering the whole syllabus. Student will attempt any 10 parts in about 25-30 words each. Each part shall carry 2 marks (total 20 marks).
2. A total of eight questions will be set out of the whole syllabus, at least 2 from each unit and the candidates will attempt 4 questions selecting one from each unit, in addition to the compulsory question at serial number I.

**(b) PRACTICAL** **Marks: 60**

(i) Familiarization with Spatial Analysis Software.
(ii) Spatial analysis: Query, Identifying, measuring, query by location/attribute
(iii) Measuring Central tendency; Mean Centre, Weighted Mean Centre, Median Centre, Standard distance, Standard Deviation Ellipse
(iv) Quadrat Analysis, Nearest Neighbour statistics
(v) Spatial Autocorrelation of Point features; Geary's Ratio, Moran's I Index
(vi) Network analysis
(vii) Vector Analysis; Buffering: Point, Line, Area buffers
(viii) Overlay
(ix) Surface Analysis; Digital Elevation Model (DEM) and Digital Terrain Model (DTM), Contouring, Hill Shading and Perspective View (3D viewing), Slope and Aspect Analysis.

**NOTE:** The practical examination shall be conducted by a team of three examiners, including the internal, Chairperson of the Department and one additional faculty member teaching the course.

**(c) INTERNAL ASSESSMENT** **Marks: 40**

Internal assessment shall be based on Written Test, Snap Test, Participation in Class discussion, Term Paper and Attendance as prescribed by the university.

**BOOKS RECOMMENDED**

**ESSENTIAL READINGS**

Chang, Kang-tsung : *Introduction to Geographic Information Systems,* Sixth Edition, Tata McGraw-Hill, 2011.

| | | |
|---|---|---|
| Chou, Yue-Hong | : | *Exploring Spatial Analysis in Geographic Information Systems,* OnWord Press, 1997. |
| Laurini, R. & Thomson, D. | : | *Fundamentals of Spatial Information Systems*, Academic Press, 1994. |
| Wong, David W.S. & Jay Lee | : | Statistical Analysis of Geographic Information With ArcView GIS and ArcGIS, John Wiley and Sons, 2005. |


**FURTHER READINGS**


| | | |
|---|---|---|
| Burrough, P.A. & Macdoneli, R.A. | : | *Principles of Geographic Information Systems,* Oxford University Press, 2000. |
| Demers, M.N. | : | *Fundamentals of geographic Information Systems*, 2nd Edition, John Wiley and sons, 2000. |
| Muralikrishna, I.V. | : | *Remote Sensing Applications and Geographic Information Systems Recent Trends*, Tata McGraw Hill, 1992. |
| Rashid, S.K. | : | Remote Sensing in Geography, Manak Publications, 1993. |
| Rees, W.G. | : | Physical Principles of Remote Sensing, Cambridge University Press, 1990. |


**PEDAGOGY:**

Students may be asked to prepare a report on spatial analysis applications using external and self generated databases. Students will be encouraged to ask questions and participate in classroom discussions.

## PAPER IX: RESEARCH METHODOLOGY AND REPORT WRITING

**Marks: 200**

**(a) THEORY** **Marks: 100**
**Time: 3 Hours**

**OBJECTIVES:**

The aim of this paper is to train the students about the various aspects of research writing so that they are in a position to write research reports independently.

**COURSE CONTENT:**

**UNIT-I**

i. Meaning, objectives and Types of Research, Research approaches and significance of research.
ii. The research process: the broad problem area, preliminary data collection, problem, selection and definition, theoretical framework, hypothesis development and elements of research design.
iii. Experimental design: the laboratory experiment, variables, validity, types of experimental designs.

**UNIT-II**

- Data collection, Measurement, processing and analysis: collection, processing and analysis; measurement in research, operational definition,
- Measurement scales, scaling; scaling techniques, reliability and validity;
- Sources of data; data collection methods: interviewing, questionnaires, other methods of data collection; review of statistical data analysis.

**UNIT-III**

- Sampling: Need and purpose of sampling, population and sample, population frame, sampling with and without replacement, population parameters.
- Sampling theory–sampling distributions, parameter estimation, hypothesis testing. Sampling designs – probability and non-probability sampling

**UNIT-IV**

- Report writing: the research proposal, the report, integral parts of the report, steps involved in report writing,
- Types of reports, oral presentation, conclusions.

**NOTE:**

1. A compulsory question containing 15 short answer type questions shall be set covering the whole syllabus. Student will attempt any 10 parts in about 25-30 words each. Each part shall carry 2 marks (total 20 marks).

2. A total of eight questions will be set out of the whole syllabus, at least 2 from each unit; and the candidates will attempt 4 questions selecting one from each unit, in addition to the compulsory question at serial number I.

**(b) PRACTICAL** **Marks: 60**

(i) Data generation from Topographical Maps, Aerial Photographs and Satellite Imageries and its analysis.
(ii) Data collection using a schedule and a questionnaire on a given problem.
(iii) Sampling exercises from the given spatial data.
(iv) Preparation of a questionnaire on a given problem.
(v) Exercises on Data Classification, Tabulation and Mapping.
(vi) Preparation of a report on a problem using RS & GIS.

**NOTE:** The practical examination shall be conducted by a team of three examiners, including the internal, Chairperson of the Department and one additional faculty member teaching the course.
Practical record book should be submitted one week before the commencement of theory exams.

**(c) INTERNAL ASSESSMENT** **Marks: 40**

Internal assessment shall be based on Written Test, Snap Test, Participation in Class discussion, Term Paper and Attendance as prescribed by the university.

**BOOKS RECOMMENDED**

**ESSENTIAL READINGS**

Ackoff, Russell L.(1961) : The Design of Social Research, Chicago; University of Chicago Press.

| | | |
|---|---|---|
| Bailey, Kenneth D.(1978) | : | Methods of Social Research, New York |
| Berdie,Douglas R and Anderson, John F. | : | Questionnaires; Design and Use, Metuchen NJ. The Scarecrow Press, Inc. |
| Creswell, John W. (2008) | : | Research Design; Qualitative, Quantitative and Mixed Methods Approach, SAGE Publications, Los Angeles. |
| Kothari, C.R.(2008) | : | Research Methodology, Methods & Techniques New Age International Publisher, N. Delhi. |
| Kumar Ranjit (2010) | : | Research Methodology A step-by-step Guide for Beginners, SAGE Publications, Ltd. London (Third Edition). |
| Misra, H.N and V.P.Singh (1998) | : | Research Methodology in Geography, Social, Spatial and Policy Dimensions, Rawat Publications, N. Delhi. |
| Smith, H.W (1975) | : | Strategies of Social Research The Methodological Imagination, Prentice-Hall, Inc., Englewood Cliffs, New Jersey. |


**FURTHER READINGS**


| | | |
|---|---|---|
| Bennet P. Lientz & Kathryn P. (1995) | : | Project Management for the 21st Century, Academic Press, California. |
| Earickson, R, and Harlin, J. (1994). | : | Geographic Measurement & Quantitative Analysis, Macmillan, New York. |
| W.E. Huxold & AG.Lerinsons Aronoft.S. (1989): | | Managing Geographic Information Projects. |


**PEDAGOGY:**

Assignments pertaining to a relevant problem shall be given to students to apply the various aspects of research.

# SEMESTER IV

**PROJECT REPORT** **Marks: 600**

| | |
|---|---|
| (a) Mid – Term Appraisal of the Project Report | Marks: 100 |
| (b) Evaluation of the Project Report | Marks: 400 |
| (c) Presentation & Viva Voce (50+50) | Marks: 100 |

**NOTE**:

(i) Mid-term appraisal of the project report of each student shall be done by the board of the four examiners consisting of the supervisor, chairman, and the two senior faculty members of the department.

(ii) Mid-term appraisal of the project report shall have two-components: (i) presentation on project proposal including the research objectives, methodology and the chapter scheme, after the three weeks of assigning the project title, (ii) presentation of data analysis and tentative findings, after the six weeks of assigning the project title.

(iii) In case of project report, the Board of Control shall assign a candidate to a faculty member/s for supervision of his/her report on an approved topic. The student shall prepare three copies of his/her project report and submit in the concerned department. The report shall have to be submitted ten days before the commencement of the theory examination of the said semester. The project report shall incorporate the findings arrived at on the basis of the data/information collected from the field and processed, analyzed and mapped in the GIS lab.

For the evaluation of project report, three examiners, consisting of supervisor, chairperson of the department, and one more faculty member teaching the course, shall evaluate the field report, submitted by the student, independently and student shall be awarded the mean score of the three evaluations. The evaluation process shall have the two-components: (i) evaluation of report, (ii) presentation of data analysis and main findings before the board of examiners and viva voce.

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